

Abstract Algebra II Homework 6

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Let L be the splitting field of f over K , and let $G = \text{Gal}(L/K)$. By its action on the four distinct roots u_1, u_2, u_3, u_4 , we can view G as a subgroup of the symmetric group S_4 .

The given elements α, β , and γ are the roots of the resolvent cubic of f . Since K is a field of characteristic zero, the field extension $K(\alpha, \beta, \gamma)/K$ is Galois, which means:

$$\text{Aut}(K(\alpha, \beta, \gamma)/K) = \text{Gal}(K(\alpha, \beta, \gamma)/K)$$

The group $G \leq S_4$ acts on the set $X = \{\alpha, \beta, \gamma\}$ by permuting the indices of the roots u_i . This action defines a natural group homomorphism from G to the symmetric group on three letters:

$$\phi : G \rightarrow S_3$$

The image of ϕ is exactly the Galois group of $K(\alpha, \beta, \gamma)$ over K . To find the structure of this group, we must determine the kernel of ϕ , which consists of the permutations in G that fix α, β , and γ simultaneously.

Evaluating the stabilizer of each element in S_4 :

- The stabilizer of α is a dihedral group of order 8: $D_8^{(1)} = \langle (12), (1324) \rangle$.
- The stabilizer of β is $D_8^{(2)} = \langle (13), (1234) \rangle$.
- The stabilizer of γ is $D_8^{(3)} = \langle (14), (1243) \rangle$.

The intersection of these three stabilizers is the normal Klein four-group in S_4 :

$$V_4 = \{\text{id}, (12)(34), (13)(24), (14)(23)\}$$

Therefore, the kernel of the homomorphism ϕ is $G \cap V_4$. By the First Isomorphism Theorem, the Galois group of the resolvent cubic is isomorphic to the quotient:

$$\text{Aut}(K(\alpha, \beta, \gamma)/K) \cong G/(G \cap V_4)$$

Because $V_4 \triangleleft S_4$ and $S_4/V_4 \cong S_3$, the structure of $\text{Aut}(K(\alpha, \beta, \gamma)/K)$ is isomorphic to a subgroup of S_3 . The exact structure depends on the specific Galois group G of the original quartic f :

- If $G = S_4$, then $\text{Aut}(K(\alpha, \beta, \gamma)/K) \cong S_3$.
- If $G = A_4$, then $\text{Aut}(K(\alpha, \beta, \gamma)/K) \cong \mathbb{Z}/3\mathbb{Z}$.
- If $G = D_8$ or $G = \mathbb{Z}/4\mathbb{Z}$, then $\text{Aut}(K(\alpha, \beta, \gamma)/K) \cong \mathbb{Z}/2\mathbb{Z}$.
- If $G = V_4$, then $\text{Aut}(K(\alpha, \beta, \gamma)/K)$ is the trivial group.

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(a)

By the rational root theorem, the possible rational roots are ± 1 . But $P(1) = P(-1) = -28 \neq 0$, so we know $P(t)$ has no linear factors over $\mathbb{Q}[t]$.

Assume that $P(t)$ factors into two monic quadratics:

$$\begin{aligned} P(t) &= (x^2 + ax + b)(x^2 + cx + d) \\ &= x^4 + (a + c)x^3 + (b + d + ac)x^2 + (ad + bc)x + bd \end{aligned}$$

By comparing the coefficients, we know:

$$\begin{aligned} a + c &= 0 \\ b + d + ac &= -30 \\ ad + bc &= 0 \\ bd &= 1 \end{aligned}$$

If $a = c = 0$, then $b + d = -30, bd = 1$. b, d are the roots for $y^2 + 30t + 1$, clearly, $b, d \notin \mathbb{Q}$. If $c = -a \neq 0$, then $b = d$, and $bd = 1$ implies $b = d = \pm 1$. In either cases, a can't be a rational number.

Therefore no factorization for $P(t)$ exists in $\mathbb{Q}[t]$, so $P(t)$ is irreducible over $\mathbb{Q}$.

(b)

The all roots of $P(t) = 0$ are $\pm 2\sqrt{2} \pm \sqrt{7}$. By grouping the roots, there are 3 possible factorization (with two monic quadratics factors):

$$\begin{aligned} P(t) &= (t^2 + 4\sqrt{2}t + 1)(t^2 - 4\sqrt{2}t + 1) \\ &= (t^2 + 2\sqrt{7}t - 1)(t^2 - 2\sqrt{7}t - 1) \\ &= (t^2 - 15 + 4\sqrt{14})(t^2 - 15 - 4\sqrt{14}) \end{aligned}$$

We can find that for every prime p (except 2 and 7), at least one of 2, 7, 14 is a quadratic residue, because when we use Legendre symbols:

$$\left(\frac{2}{p}\right) \left(\frac{7}{p}\right) = \left(\frac{14}{p}\right)$$

If both $\left(\frac{2}{p}\right) = \left(\frac{7}{p}\right) = -1$, then $\left(\frac{14}{p}\right) = 1$.

Therefore, if $\exists s \in \mathbb{F}_p$ ($p \neq 2, 7$) such that $s^2 = 2$, then we can write $P(t) = (t^2 + 4st + 1)(t^2 - 4st + 1)$. If $\exists u \in \mathbb{F}_p$ ($p \neq 2, 7$) such that $u^2 = 7$, then we can write $P(t) = (t^2 + 2ut - 1)(t^2 - 2ut - 1)$. If $\exists v \in \mathbb{F}_p$ ($p \neq 2, 7$) such that $v^2 = 14$, then we can write $P(t) = (t^2 - 15 + 4v)(t^2 - 15 - 4v)$. Here, $P(t)$ is considered under $\mathbb{F}_p[t]$ for some prime $p \neq 2, 7$.

Let's consider the cases when $p = 2, 7$. When $p = 2$, $P(t) = x^4 + 1 = (x + 1)^4 \in \mathbb{F}_2[t]$. When $p = 7$, $P(t) = x^4 + 5x^2 + 1 = (x^2 - 1)^2 \in \mathbb{F}_7[t]$.

Hence, $P(t) \bmod p \in \mathbb{F}_p[t]$ is reducible for all prime numbers p .

(c)

The four roots of $f(x)$ are $\pm 2\sqrt{2} \pm \sqrt{7}$. Since all roots can be generated by adjoining $\sqrt{2}$ and $\sqrt{7}$ to $\mathbb{Q}$, the splitting field is:

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{7})$$

Since $\sqrt{2}$ and $\sqrt{7}$ are square roots of distinct prime numbers, the degree of the extension is $[K : \mathbb{Q}] = 4$. The Galois group $G = \text{Gal}(K/\mathbb{Q})$ has order 4. Because K is generated by two independent square roots, G is isomorphic to the Klein four-group, $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

The group G consists of four automorphisms, completely determined by their action on the generators $\sqrt{2}$ and $\sqrt{7}$:

- $id: \sqrt{2} \mapsto \sqrt{2}, \quad \sqrt{7} \mapsto \sqrt{7}$
- $\sigma: \sqrt{2} \mapsto -\sqrt{2}, \quad \sqrt{7} \mapsto \sqrt{7}$
- $\tau: \sqrt{2} \mapsto \sqrt{2}, \quad \sqrt{7} \mapsto -\sqrt{7}$
- $\sigma\tau: \sqrt{2} \mapsto -\sqrt{2}, \quad \sqrt{7} \mapsto -\sqrt{7}$

By the Fundamental Theorem of Galois Theory, the subgroups of G correspond to the intermediate fields of $K/\mathbb{Q}$. The proper, non-trivial subgroups of G are $\langle \sigma \rangle$, $\langle \tau \rangle$, and $\langle \sigma\tau \rangle$, all of which have order 2 (and thus index 2 in G). We find their corresponding fixed fields:

- $K^{\langle \sigma \rangle}$: The automorphism σ fixes $\sqrt{7}$. Thus, the intermediate field is $\mathbb{Q}(\sqrt{7})$.
- $K^{\langle \tau \rangle}$: The automorphism τ fixes $\sqrt{2}$. Thus, the intermediate field is $\mathbb{Q}(\sqrt{2})$.
- $K^{\langle \sigma\tau \rangle}$: The automorphism $\sigma\tau$ negates both $\sqrt{2}$ and $\sqrt{7}$. Their product, $\sqrt{2}\sqrt{7} = \sqrt{14}$, is mapped to $(-\sqrt{2})(-\sqrt{7}) = \sqrt{14}$. Thus, it fixes $\sqrt{14}$, giving the intermediate field $\mathbb{Q}(\sqrt{14})$.

